EE4-45
EE9-AML3-04
EE9-CS7-21
EE9-SO22

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

WAVELETS, REPRESENTATION LEARNING AND THEIR APPLICATIONS

Wednesday, 28 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P.L. Dragotti
Second Marker(s) : A. Manikas

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Special Information for the Invigilators: NONE

Information for Candidates:

Sub-sampling by an integer N

$$x_{\downarrow N}[n] \longleftrightarrow \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j(\omega-2\pi k)/N}) = \frac{1}{N}\sum_{k=0}^{N-1} X(W_N^k z^{1/N}),$$

where

$$W_N^k = e^{-j2\pi k/N}.$$

Dual Basis:
Given a basis $\{\varphi_i(t)\}_{i\in\mathbb{Z}}$, the dual basis is given by the set of elements $\{\tilde{\varphi}_i(t)\}_{i\in\mathbb{Z}}$ satisfying:

$$\langle \varphi_i(t), \tilde{\varphi}_j(t)\rangle = \delta_{i,j}.$$

Haar scaling and wavelet functions
The Haar scaling function is defined as:

$$\varphi(t) = \begin{cases} 1 & 0 \le t < 1 \\ \\ 0 & \text{otherwise} \end{cases}$$

The Haar wavelet is defined as:

$$\psi(t) = \begin{cases} 1 & 0 \le t < 1/2 \\ \\ -1 & 1/2 \le t < 1 \\ \\ 0 & \text{otherwise} \end{cases}$$

The Questions

1. Assume $\psi(t)$ is the Haar wavelet. Given a signal $f(t) \in L_2(\mathbb{R})$, we know that we can express it as follows:

$$f(t) = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \alpha_{m,n} \psi_{m,n}(t),$$

where $\psi_{m,n}(t) = 2^{-m/2}\psi(2^{-m}t - n)$.

(a) Assume

$$f(t) = \begin{cases} 1 & 0 \leq t < 1 \\ \\ 0 & \text{otherwise.} \end{cases}$$

Derive the expansion coefficients $\alpha_{m,n}$ for this case. [5]

(b) Verify that $\sum_n \sum_m |\alpha_{m,n}|^2 = \|f(t)\|^2$. [5]

(c) In a multiresolution analysis with scaling function $\varphi(t)$ satisfying the orthogonality condition $\langle \varphi(t), \varphi(t-n) \rangle = \delta_n$, consider the two scale equation

$$\varphi(t) = \sqrt{2} \sum_n g_0[n] \varphi(2t - n).$$

Assume that $0 < |\hat{\varphi}(0)| < +\infty$ where $\hat{\varphi}(\cdot)$ is the Fourier transform of $\varphi(t)$. Moreover, assume that $\hat{\varphi}(\omega)$ is continuous in $\omega = 0$.

i. Derive the expression of the two-scale equation in Fourier domain [5]

ii. Show that $\sum_n g_0[n] = \sqrt{2}$ [5]

iii. Show that $\sum_n g_0[2n] = \sum_n g_0[2n+1]$ [5]

2. Consider the two systems shown in Fig. 2a and Fig. 2b.

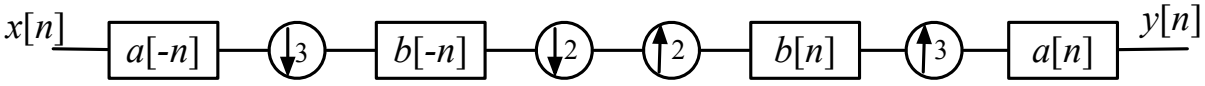


Figure 2a: First multirate system.

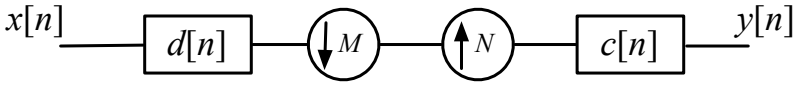


Figure 2b: Second multirate system.

(a) Find the filters $c[n]$ and $d[n]$ and integers M and N so that the system in Fig. 2b has input-output relationship equivalent to the system in Fig 2a. (Either a time domain or a z domain answer is acceptable.)

[5]

(b) Given that the filter $b[n]$ satisfies $\langle b[n], b[n-2k] \rangle = \delta_k$, find (and prove) a sufficient condition on $a[n]$ such that the mapping from x to y is idempotent. Idempotent means that if the mapping was applied again to the sequence $y[n]$, the output would stay $y[n]$.

[5]

(c) Consider the system

$$y = HD_4GD_2FD_3x$$

where G, H and F are the infinite matrix representations for filtering and D_k is the matrix representation of downsampling by k with $k = 2, 3, 4$.

i. Draw a block diagram of the system.

[5]

ii. Derive an equivalent system consisting of a single filter block and a single downsampling matrix. Write the z-transform of the equivalent system as a function of $G(z)$, $H(z)$ and $F(z)$

[5]

Question continues on next page.

(d) Consider now a filter bank specified by the following signal equations:

$$\begin{aligned} y_0 &= D_2GD_2Gx \\ y_1 &= D_2HD_2Gx \\ y_2 &= D_2Hx. \end{aligned}$$

i. Draw the equivalent single-level three-channel filter bank and derive the exact transfer functions of the equivalent filters assuming that g_n and h_n are the lowpass and highpass Haar filters respectively. Specifically, the z-transform of g_n is $G(z) = (1+z)/\sqrt{2}$ and $H(z) = (1-z)/\sqrt{2}$. [3]

ii. Assume that $x[n] = 1$ and ignore any boundary effect. Which of the signals $y_0[n]$,$y_1[n]$ and $y_2[n]$ is nonzero? (Justify your answer). [2]

3. Consider the two-channel filter bank of Figure 3a.

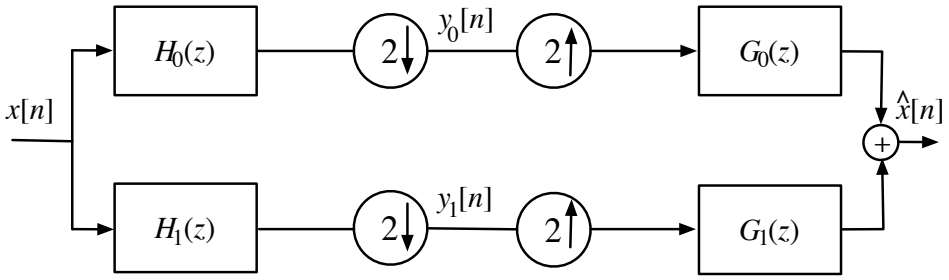


Figure 3a: Two-channel filter bank.

(a) Express $\hat{X}(z)$ as a function of $X(z)$ and the filters. Then derive the two perfect reconstruction (PR) conditions the filters have to satisfy. [5]

(b) Assume that

$$G_0(z) = (z + 2 + z^{-1})/(2\sqrt{2}).$$

We would like $H_0(z)$ such that $P(z) = H_0(z)G_0(z)$ satisfies $P(z) + P(-z) = 2$. Assume that $H_0(z)$ is a 5-length symmetric filter: $H_0(z) = cz^2 + bz + a + bz^{-1} + cz^{-2}$.

i. Show that you can have many filters $H_0(z)$ leading to $P(z) + P(-z) = 2$. [5]

ii. To have a unique solution, impose that the filter have a zero at $z = -1$. Find the unique 5-length symmetric filter $H_0(z)$ satisfying $P(z) + P(-z) = 2$ and with a zero at $z = -1$. [5]

iii. Find the filters $H_1(z)$ and $G_1(z)$ that would lead to a perfect reconstruction filter-bank. [5]

Question continues on next page.

(c) Consider the equivalent filter

$$H_0^{(i)}(z) = \prod_{k=0}^{i-1} H_0(z^{2^k})$$

obtained by iterating filter $H_0(z)$ in the filter bank decomposition i times. Consider the function

$$\varphi^{(i)}(t) = 2^{i/2} h_0^{(i)}[n], \qquad n/2^i \leq t < (n+1)/2^i.$$

i. Can you say anything about the convergence of $\varphi(t) = \lim_{i\to\infty} \varphi^{(i)}(t)$? [3]

ii. Assume that $\varphi(t) = \lim_{i\to\infty} \varphi^{(i)}(t)$ exists. State the number of vanishing moments of the wavelet

$$\psi(t) = \sqrt{2} \sum_n (-1)^{n-1} h_0[1-n] \varphi(2t-n).$$

Justify your answer. [2]

4. Consider the interval $t \in [0,1]$ and the function

$$x(t) = \begin{cases} \sin \frac{1}{2}\pi t, & \text{for } t \in [0,1] \\ \\ 0 & \text{otherwise} \end{cases}$$

your goal is to find a polynomial of degree one that best approximates $x(t)$ over the interval $[0,1]$. Specifically, let

$$\varphi_1(t) = \begin{cases} 1, & \text{for } t \in [0,1] \\ \\ 0, & \text{otherwise} \end{cases}$$

and

$$\varphi_2(t) = \begin{cases} t, & \text{for } t \in [0,1] \\ \\ 0, & \text{otherwise,} \end{cases}$$

the degree one polynomial is given by $p(t) = \sum_{k=1}^{2} a_k \varphi_k(t)$ and your goal is to find a_1 and a_2 that minimise

$$\|e(t)\|^2 = \int_0^1 e^2(t)dt.$$

with $e(t) = x(t) - p(t)$.

(a) Use Lagrange interpolation to find a possible polynomial. Specifically, find a_1 and a_2 such that $p(0) = x(0)$ and $p(1) = x(1)$. [5]

(b) By using the projection theorem, show that this polynomial does not minimise $\|e(t)\|^2$. [5]

Question continues on next page.

(c) Denote with $V = \text{span}(\{\varphi_1(t), \varphi_2(t)\})$ the sub-space generated by $\varphi_i(t)$ with $i = 1, 2$ over the interval $t \in [0, 1]$, the minimization of $\|e(t)\|^2$ is achieved by computing the orthogonal projection of $x(t)$ onto V. Recall that this is given by:

$$x_v(t) = \sum_{i=1}^{2} \langle x(t), \tilde{\varphi}_i(t) \rangle \varphi_i(t)$$

where $\{\tilde{\varphi}_i(t)\}_{i=1}^2$ are the two dual-basis functions.

i. Since $\tilde{\varphi}_i(t) \in V$, we can write $\tilde{\varphi}_i(t) = \sum_{k=1}^{2} \alpha_{i,k} \varphi_k(t)$. Using this fact, determine the two dual-basis functions $\tilde{\varphi}_i(t)$, $i = 1, 2$. That is, find the coefficients $\alpha_{i,k}$, $i = 1, 2$; $k = 1, 2$. [4]

ii. Compute the inner products $\langle x(t), \tilde{\varphi}_i(t) \rangle$, $i = 1, 2$. [4]

iii. Write the exact expression for $x_v(t) = \sum_{i=1}^{2} \langle x(t), \tilde{\varphi}_i(t) \rangle \varphi_i(t)$. [3]

iv. Verify that the error $e_v(t) = x(t) - x_v(t)$ is orthogonal to V. [4]